Number of articles = (5 + 4 + 12) = 21.

$\therefore$ Average expenditure = ₹ $\left(\frac{14250}{21}\right)$ = ₹ $\frac{4750}{7}$ = ₹ 678.57.

Ex. 8. *13 chairs and 5 tables were bought for ₹ 8280. If the average cost of a table be ₹ 1227, what is the average cost of a chair?* **(S.S.C., 2005)**

Sol. Total cost of 5 tables = ₹ (1227 × 5) = ₹ 6135.

Total cost of 13 chairs = ₹ (8280 – 6135) = ₹ 2145.

$\therefore$ Average cost of a chair = ₹ $\left(\frac{2145}{13}\right)$ = ₹ 165.

Ex. 9. *The average of five consecutive numbers A, B, C, D and E is 48. What is the product of A and E?* **(Bank Recruitment, 2008)**

Sol. Let the numbers A, B, C, D and E be x, $(x + 1)$, $(x + 2)$, $(x + 3)$ and $(x + 4)$ respectively. Then,

$$\frac{x+(x+1)+(x+2)+(x+3)+(x+4)}{5} = 48 \Rightarrow 5x + 10 = 240 \Rightarrow 5x = 230 \Rightarrow x = 46.$$

So, A = x = 46 and $E = (x + 4) = 50$.

$\therefore$ Required product = 46 × 50 = 2300.

Ex. 10. *The average monthly expenditure of a family was ₹ 2200 during the first 3 months; ₹ 2250 during the next 4 months and ₹ 3120 during the last 5 months of a year. If the total savings during the year were ₹ 1260, find the average monthly income of the family.* **(M.A.T., 2006)**

Sol. Total yearly expenditure = ₹ (2200 × 3 + 2250 × 4 + 3120 × 5)

= ₹ (6600 + 9000 + 15600) = ₹ 31200.

Total yearly savings = ₹ 1260.

Total yearly income = ₹ (31200 + 1260) = ₹ 32460.

$\therefore$ Average monthly income = ₹ $\left(\frac{32460}{12}\right)$ = ₹ 2705.

Ex. 11. *Six persons went to a hotel for taking their meals. Five of them spent ₹ 32 each on their meals while the sixth person spent ₹ 80 more than the average expenditure of all the six. What was the total money spent by all the persons?* **(C.P.O., 2006)**

Sol. Let the average expenditure of all the six be ₹ x.

Then, $32 \times 5 + (x + 80) = 6x \Rightarrow 240 + x = 6x \Rightarrow 5x = 240 \Rightarrow x = 48$.

$\therefore$ Total money spent = $6x$ = ₹ (6 × 48) = ₹ 288.

Ex. 12. *The average age of a man and his son is 40 years. The ratio of their ages is 11 : 5 respectively. What is the son's age?* **(Bank Recruitment, 2009)**

Sol. Let the ages of the man and his son be $11x$ and $5x$ years respectively.

Then, average age = $\left(\frac{11x+5x}{2}\right)$ years = $8x$ years. $\therefore$ $8x = 40 \Rightarrow x = 5$.

Hence, son's age = $5x$ = 25 years.

Ex. 13. *Of the three numbers, second is twice the first and is also thrice the third. If the average of three numbers is 44, what is the largest number?* **(M.B.A., 2007)**

Sol. Let the third number be x. Then, second number = $3x$.

First number = $\frac{3x}{2}$.

$$\therefore\ x + 3x + \frac{3x}{2} = 44 \times 3 \Rightarrow \frac{11x}{2} = 132 \Rightarrow x = \left(\frac{132 \times 2}{11}\right) = 24.$$

Hence, largest number = $3x$ = (3 × 24) = 72.